Model RFB RASCOFLEX® Sprinkler Connections

cULus Listed, FM Approved

Product Description

RASCOFLEX® Sprinkler Connections are intended to connect a branch line to a sprinkler using a flexible stainless steel hose assembly. RASCOFLEX® Sprinkler Connections are suitable for use in suspended and hard ceiling applications such as T-Bar ceiling grids, wood, metal stud, or hat furring channel hard lid ceilings. Every package contains one (1) fully assembled stainless steel flexible sprinkler system complete with hose, branch line connection, and sprinkler connection, as well as the pre-assembled bracket assembly to attach to the ceiling.

RASCOFLEX® Sprinkler Connections are designed for use in hydraulically calculated wet, preaction, or dry sprinkler systems per NFPA 13, 13R, 13D, and FM Global Loss Prevention Data Sheets.

Installation

Please refer to RASCOFLEX® Installation Instructions.

Maintenance

RASCOFLEX® Sprinkler Connections should be inspected and the sprinkler system maintained in accordance with NFPA 25, as well as the requirements of any Authorities Having Jurisdiction.

Patents

RASCOFLEX® Sprinkler Connections may be covered by one or more of the following US Patent Nos. 10,173,088 and 10,328,296.

Listings and Approvals

FM Approved Class No. 1637 (FM)
UL Listed and UL Certified for Canada to ANSI/UL 2443 (cULus)

Technical Data **Table A**

Maximum Working Pressure		FM: 200 psi (13.8 bar) UL: 175 psi (12.1 bar)
Maximum Working Temperature		300°F (149°C)
Connections	Inlet/Branch Line	1” NPT
	Outlet/Reducer	1/2” or 3/4” NPT
Minimum Allowable Bending Radius		UL: 3” (76 mm) FM: 7” (178 mm)
Maximum Number of Bends		See Friction Loss Chart
Maximum K-Factor	1/2” Outlet	5.6 (80 metric)
	3/4” Outlet	14.0 (200 metric)



RASCOFLEX® Sprinkler Connections

Ordering Information

Specify:

Model Name
- Model RFB

Bracket Height
- Standard (B1)
- Tall (B2)

Bracket Length
- 24” (610 mm) standard
- 48” (1220 mm) optional

Nominal Hose Length
- 24” (610 mm)
- 31” (790 mm)
- 40” (1015 mm)
- 48” (1220 mm)
- 60” 1525 mm)
- 72” (1830 mm)

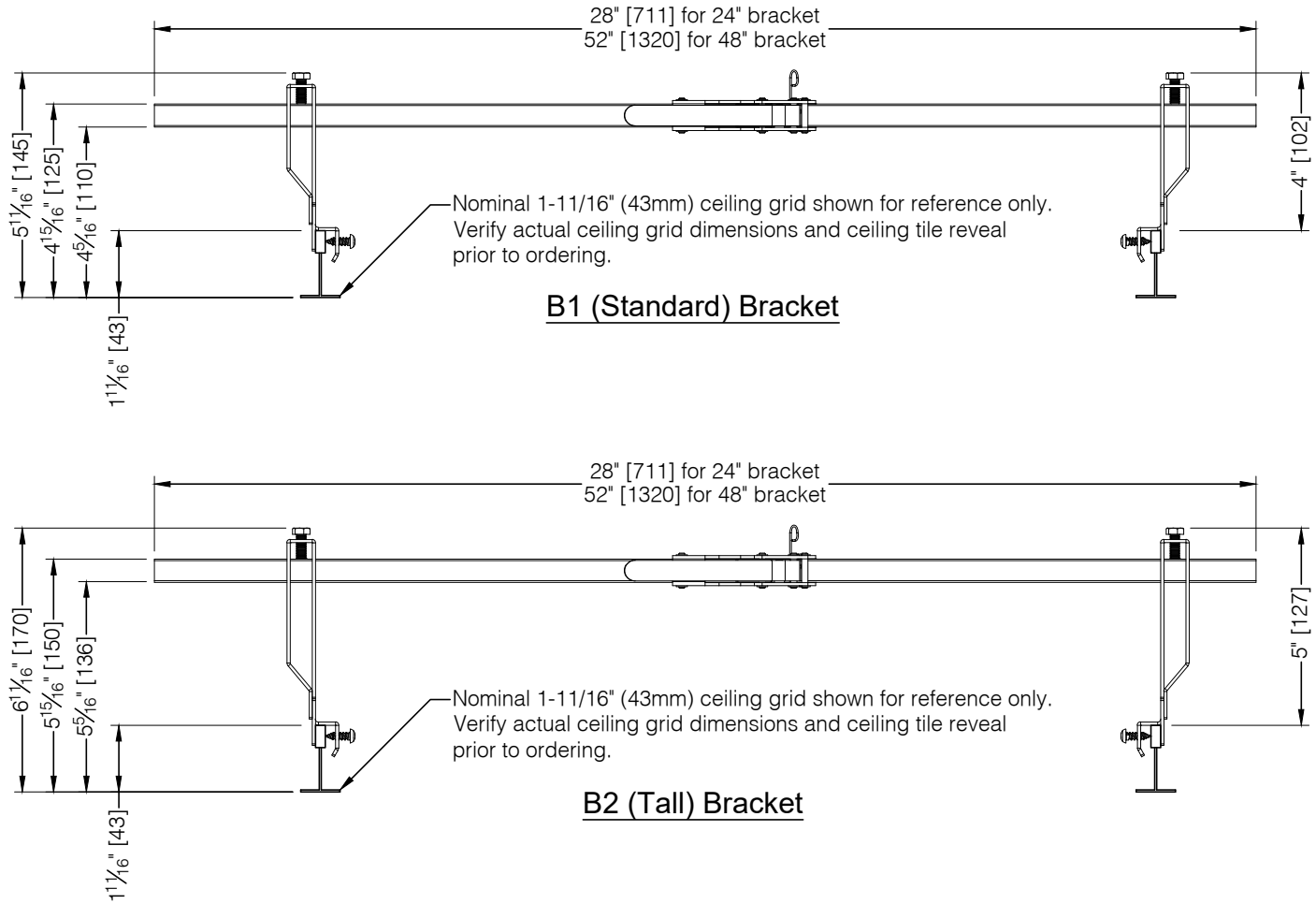
Reducer Outlet: 1/2” NPT or 3/4” NPT

Reducer Type
Standard:
- 7” (178 mm) straight

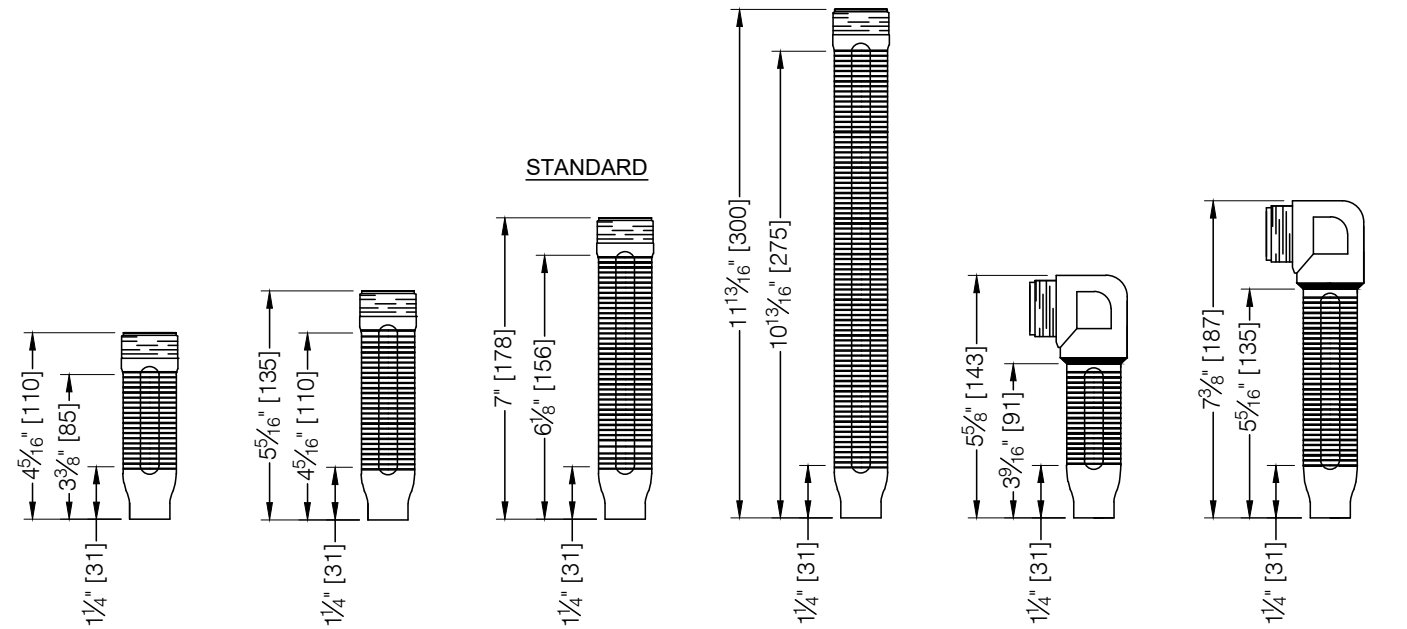
Optional:
- 4-5/16” (110 mm) straight
- 5-5/16” (135 mm) straight
- 11-3/4” (300 mm) straight
- 5-5/8” (143 mm) elbow
- 7-3/8” (187 mm) elbow

Accessories
- See Table F

Bracket Dimensions **Figure 1**

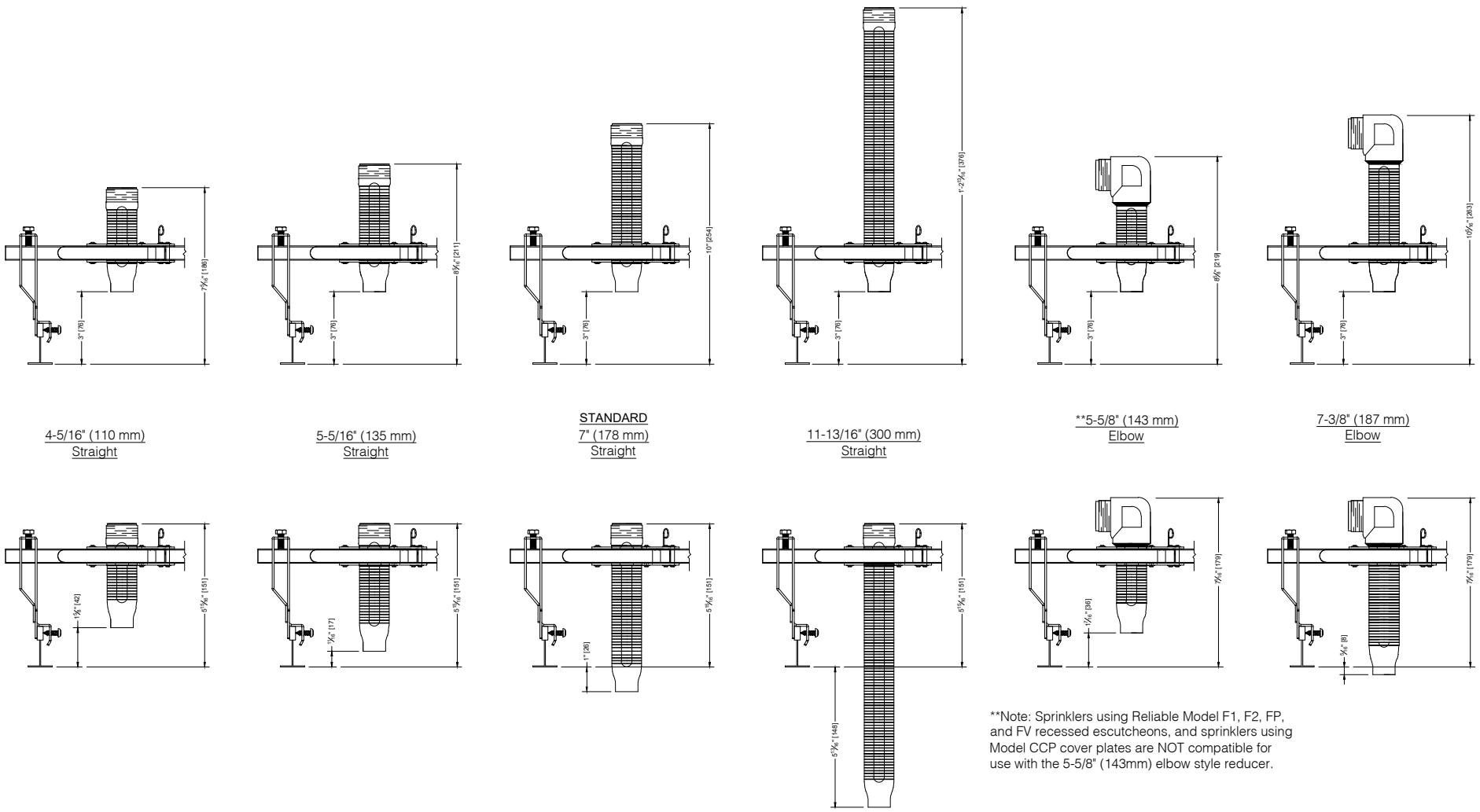


Reducer Dimensions **Figure 2**



Installation Dimensions - Minimum and Maximum Distance from Ends of Reducer to Ceiling at Fully Recessed and Fully Extended Positions: Standard (B1) Bracket

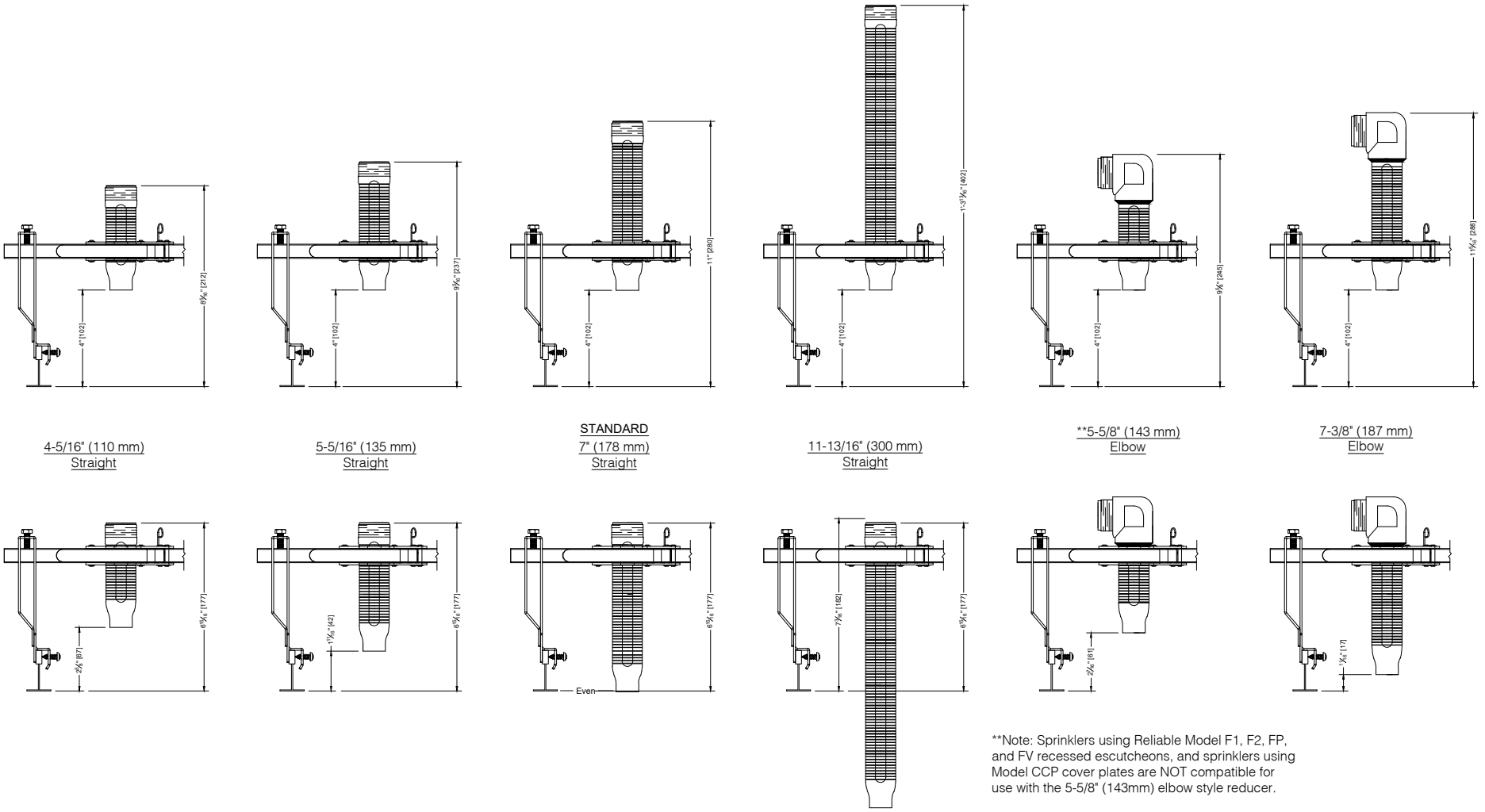
Figure 3A



**Note: Sprinklers using Reliable Model F1, F2, FP, and FV recessed escutcheons, and sprinklers using Model CCP cover plates are NOT compatible for use with the 5-5/8" (143mm) elbow style reducer.

Installation Dimensions - Minimum and Maximum Distance from Ends of Reducer to Ceiling at Fully Recessed and Fully Extended Positions: Tall (B1) Bracket

Figure 3B



4-5/16" (110 mm) Straight

5-5/16" (135 mm) Straight

STANDARD
7" (178 mm) Straight

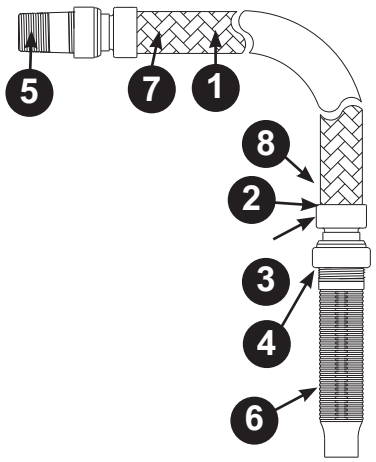
11-13/16" (300 mm) Straight

**5-5/8" (143 mm) Elbow

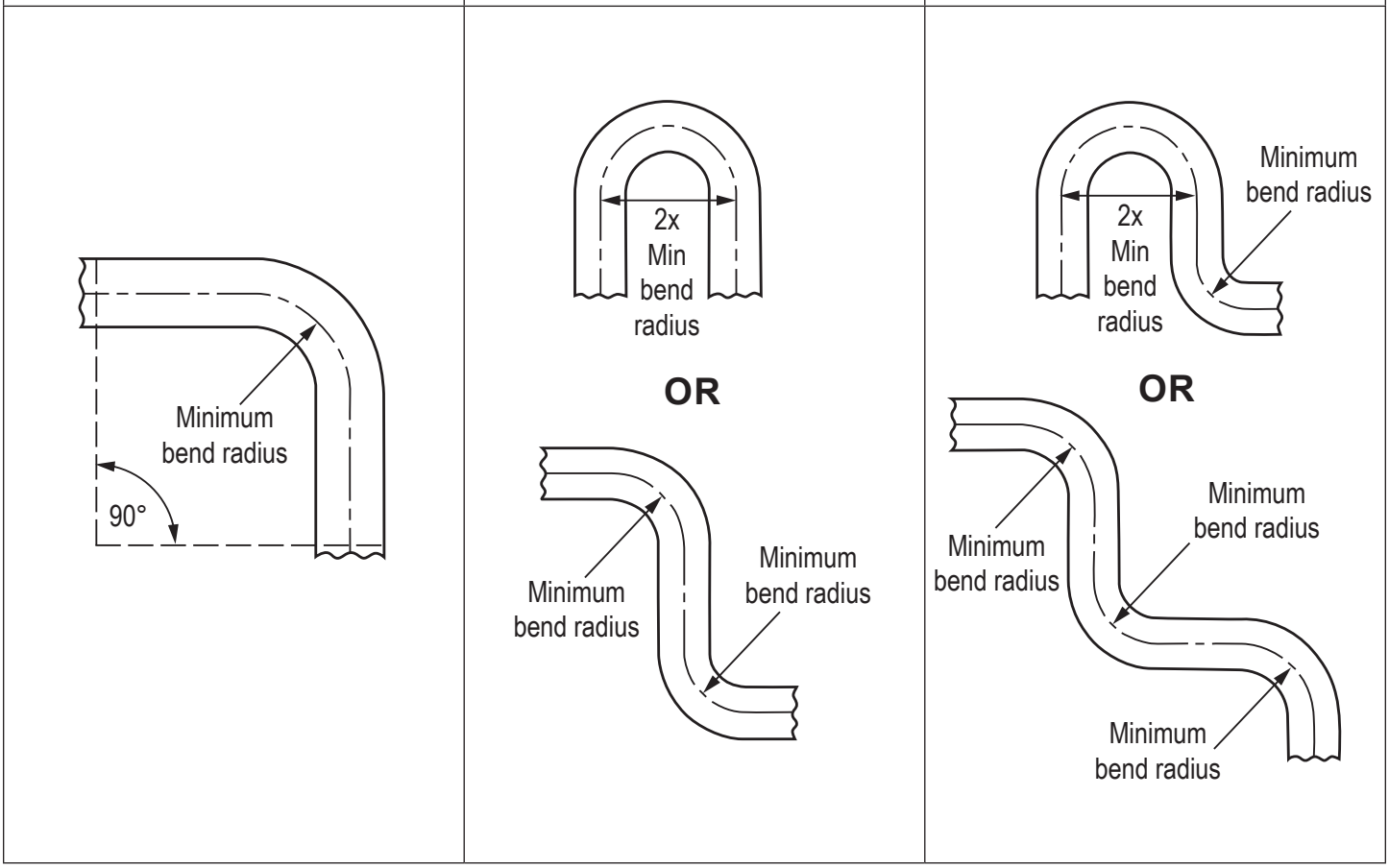
7-3/8" (187 mm) Elbow

**Note: Sprinklers using Reliable Model F1, F2, FP, and FV recessed escutcheons, and sprinklers using Model CCP cover plates are NOT compatible for use with the 5-5/8" (143mm) elbow style reducer.

Materials			Figure 4
Number	**Item Description**	**Material**	
1	Flexible Hose/Bellow	AISI Type 304 Stainless Steel	
2	Isolation Ring	Nylon 66	
3	Gasket	EPDM	
4	Nut	Zinc Plated Carbon Steel	
5	Branch Line Nipple (1”)	Zinc Plated Carbon Steel	
6	Reducer	Zinc Plated Carbon Steel	
7	Braid	AISI Type 304 Stainless Steel	
8	Welded Collar Fitting	AISI Type 304 Stainless Steel	
-	Bar Stock	Zinc Plated SGCC Steel	
-	Brackets: Center and Side	Zinc Plated SPCC Steel	



Flexible Hose Number of Bends — **Figure 5**



Do NOT install the RASCOFLEX™ straight. Some flexibility in the form of an allowable bend (or bends) must be provided.

Painting

All Reliable RASCOFlex products may be painted provided the paint is compatible with the following materials of construction:

- Stainless steel
- Zinc-plated carbon steel
- Ductile iron

Care should be taken to ensure that the sprinkler and associated escutcheon or cover plate are not painted. Additionally, care should be taken to avoid painting over any manufacturer or product identifying markings or tags on the flexible drop. No paint of any kind should be applied to threads or components / surfaces that comprise a pressure-containing seal.

cULus Friction Loss Data **Table D**

Nominal Length of Flexible Hose in (mm)	Reducer		Maximum Sprinkler K-Factor gpm/psi$^{1/2}$ (lpm/bar$^{1/2}$)	Maximum Number of 90° Bends at 3" (76mm) Bend Radius	Equivalent Length of 1" (33.7mm) Sch. 40 Pipe (C=120), ft (m)
	NPT Threads	Type			
24 (610)	1/2	Straight	5.6 (80)	2	10 (3)
	3/4	Straight	14.0 (200)	2	13 (4)
31 (790)	1/2	Straight	5.6 (80)	3	14 (4.3)
	3/4	Straight	14.0 (200)	3	16 (4.9)
40 (1015)	1/2	Straight	5.6 (80)	4	21 (6.4)
	3/4	Straight	14.0 (200)	4	23 (7)
48 (1220)	1/2	Straight	5.6 (80)	4	24 (7.3)
	3/4	Straight	14.0 (200)	4	26 (7.9)
60 (1525)	1/2	Straight	5.6 (80)	4	25 (7.6)
	3/4	Straight	14.0 (200)	4	30 (9.1)
72 (1830)	1/2	Straight	5.6 (80)	5	36 (11)
	3/4	Straight	14.0 (200)	5	33 (10.1)

UL Notes:

1. Data is based on standard 7" (178mm) straight reducer.
2. Sprinkler K-Factor: 5.6 (80 metric) for 1/2-inch reducer and 14.0 (200 metric) for 3/4-inch reducer.
3. RASCOFLEX® Sprinkler Connections have been tested and approved by Underwriter's Laboratories, Inc. for use in wet, preaction, and dry sprinkler systems per NFPA 13, 13D, 13R and UL2443.

FM APPROVED

FM Friction Loss Data

Table E

Nominal Length of Flexible Hose in (mm)	Reducer		Maximum Sprinkler K-Factor gpm/psi$^{1/2}$ (lpm/bar$^{1/2}$)	Maximum Number of 90° Bends at 7" 178mm) Bend Radius	Equivalent Length of 1" (33.7mm) Sch. 40 Pipe (C=120), ft (m)
	NPT Threads	Type			
24 (610)	1/2	Straight	5.6 (80)	1	9.7 (2.9)
	1/2	90° Elbow	5.6 (80)	0	11.5 (3.5)
	3/4	Straight	8.0 (115)	1	9.9 (3)
			11.2 (160)	1	9.8 (2.9)
			14.0 (200)	1	9.6 (2.9)
	3/4	90° Elbow	8.0 (115)	0	10.2 (3.1)
			11.2 (160)	0	10 (3)
			14.0 (200)	0	9.8 (2.9)
31 (790)	1/2	Straight	5.6 (80)	2	12.4 (3.8)
	1/2	90° Elbow	5.6 (80)	2	15.8 (4.8)
	3/4	Straight	8.0 (115)	2	13.7 (4.1)
			11.2 (160)	2	12.9 (3.9)
			14.0 (200)	2	12.2 (3.7)
	3/4	90° Elbow	8.0 (115)	2	14.5 (4.4)
			11.2 (160)	2	13.7 (4.1)
			14.0 (200)	2	13 (3.9)
40 (1015)	1/2	Straight	5.6 (80)	2	15.9 (4.8)
	1/2	90° Elbow	5.6 (80)	2	21.6 (6.6)
	3/4	Straight	8.0 (115)	2	18.5 (5.6)
			11.2 (160)	2	17.4 (5.3)
			14.0 (200)	2	16.3 (4.9)
	3/4	90° Elbow	8.0 (115)	2	20 (6)
			11.2 (160)	2	18.9 (5.7)
			14.0 (200)	2	20 (6)
48 (1220)	1/2	Straight	5.6 (80)	3	19.0 (5.8)
	1/2	90° Elbow	5.6 (80)	3	25.9 (7.9)
	3/4	Straight	8.0 (115)	3	22.7 (6.9)
			11.2 (160)	3	21.5 (6.5)
			14.0 (200)	3	20.5 (6.2)
	3/4	90° Elbow	8.0 (115)	3	24.8 (7.5)
			11.2 (160)	3	23.6 (7.2)
			14.0 (200)	3	22.6 (6.8)

FM Friction Loss Data (cont.)

Table E

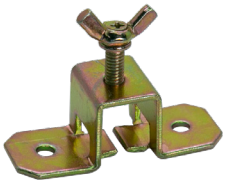
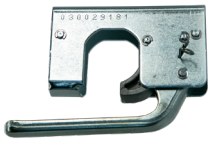
Nominal Length of Flexible Hose in (mm)	Reducer		Maximum Sprinkler K-Factor gpm/psi$^{1/2}$ (lpm/bar$^{1/2}$)	Maximum Number of 90° Bends at 7" (178mm) Bend Radius	Equivalent Length of 1" (33.7mm) Sch. 40 Pipe (C=120), ft (m)
	NPT Threads	Type			
60 (1525)	1/2	Straight	5.6 (80)	4	23.7 (7.2)
	1/2	90° Elbow	5.6 (80)	4	33.1 (10)
	3/4	Straight	8.0 (115)	4	29.1 (8.8)
			11.2 (160)	4	28 (8.5)
			14.0 (200)	4	27 (8.2)
	3/4	90° Elbow	8.0 (115)	4	32.2 (9.8)
			11.2 (160)	4	31.1 (9.5)
			14.0 (200)	4	30 (9.1)
72 (1830)	1/2	Straight	5.6 (80)	4	28.4 (8.6)
	1/2	90° Elbow	5.6 (80)	4	40.4 (12.3)
	3/4	Straight	8.0 (115)	4	35.5 (10.8)
			11.2 (160)	4	34.3 (10.4)
			14.0 (200)	4	33.2 (10.1)
	3/4	90° Elbow	8.0 (115)	4	39.5 (12)
			11.2 (160)	4	38.3 (11.6)
			14.0 (200)	4	37.2 (11.3)

FM Notes:

1. Data is based on standard 7" (178mm) straight reducer.
2. RASCOFLEX® Sprinkler Connections have been tested and approved by FM Approvals for use in wet, preaction, and dry sprinkler systems per FM data sheets 2-0, 2-5, and 2-8 per FM1637.
3. Maximum sprinkler K-Factor: 5.6 (80 metric) for 1/2-inch reducer and 14.0 (200 metric) for 3/4-inch reducer.
4. Differences in equivalent lengths are due to varying test methods, per FM 1637 standards.

Accessories List

Table F

 5-5/8” (143mm) Elbow Reducer- Short 1/2”: 7M99003303 3/4”: 7M99003305	 7-3/8” (187mm) Elbow Reducer- Long 1/2”: 7M99003302 3/4”: 7M99003304	 4-5/16” (110mm) Straight Reducer 1/2”: 7M99003306 3/4”: 7M99003325
 5-5/16” (135mm) Straight Reducer 1/2”: 7M99003307 3/4”: 7M99003326	 Replacement 7” (178mm) Standard Straight Reducer 1/2”: 7M99004856 3/4”: 7M99004869	 11-13/16” (300mm) Straight Reducer 1/2”: 7M99003309 3/4”: 7M99003328
 Hat Channel End Bracket- Short 3” (76mm) 7M99003310	 Hat Channel End Bracket- Long 3-3/4” (95mm) 7M99003311	 Metal Stud End Bracket-Short 1-1/2” (38mm) 7M99003312
 Metal Stud End Bracket- Long 2-1/16” (53mm) 7M99003313	 T-Bar End Bracket- Short 2-5/8” (68mm) 7M99003314	 T-Bar End Bracket- Long 4-1/8” (105mm) 7M99003316
 Wood Beam Stud End Bracket 7M99003317	 Replacement Center Bracket 7M99003321	 3” (76 mm) Bend Radius Indicator 7M99004179
 Replacement 1” NPT Inlet Adapter 7M99003322	 #2 Square Drive Bit 7M99004539	 Replacement Gasket 7M99004319
 48” (1220mm) Long Bracket Assembly Std. Height: 7M99003301 Tall Bracket: 7M99004840		